



Induction of Labor

Objectives

By the end of this lecture students will be able to:

1. Define what IOL is.
2. List the causes of IOL.
3. Enumerate the contraindication of IOL.
4. Outline the routes of IOL



What is IOL?

It's the process of artificially initiating uterine contractions, prior to their spontaneous onset , with the intention to effect progressive effacement and dilatation of the cervix and ultimately delivery of the baby.

Indications

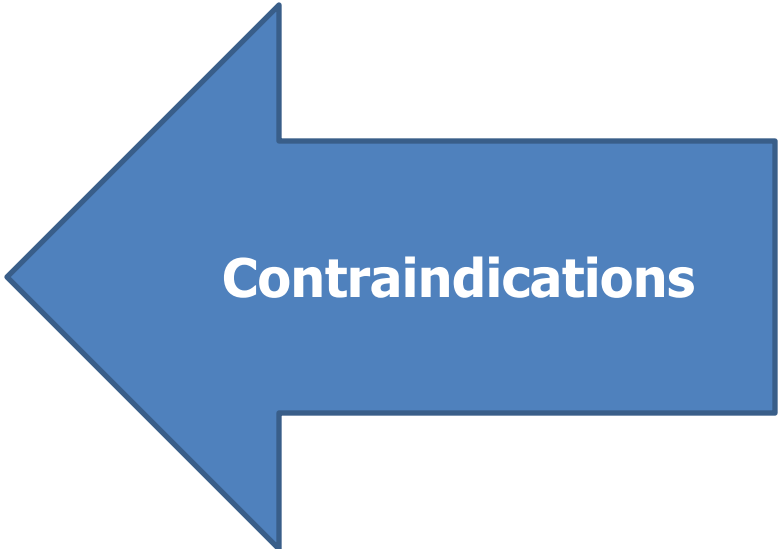
Maternal causes:

- Prolonged pregnancy
- PROM
- Maternal diseases :DM , renal diseases , autoimmune diseases, HTN
- Pregnancy related conditions:
- Pre-eclampsia
- Intrahepatic cholestasis of pregnancy.
- Recurrent APH.

Fetal causes:

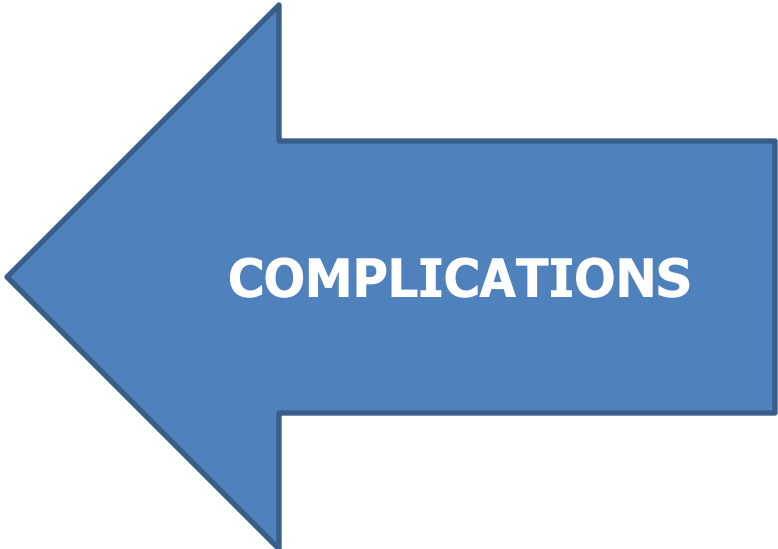
- IUGR
- IUFD
- Rh- isoimmunization

- ❖ Major degree placenta previa
- ❖ Severe IUGR & fetal compromise.
- ❖ Severe cephalopelvic disproportion
- ❖ Deteriorating maternal condition with PET or cardiac diseases.
- ❖ Transverse lie
- ❖ Breech presentation is relative contraindication to IOL.
- ❖ Prev. 2 and more C/S , myomectomy
- ❖ Prev.1 C/S is relative contraindication to IOL



Contraindications

- Increased pain and need for analgesia.
- Uterine hyperstimulation.
- Fetal distress.
- Intrauterine infections.
- Uterine rupture(rare)
- PPH
- Increased risk of operative vaginal delivery and birth trauma.
- Failure leading to C/S



COMPLICATIONS

Prerequisites

- Establish indication clearly
- Informed consent
- Confirmation of gestational age
- Assessment of fetal size , lie & presentation
- Do pelvic assessment & assesment of Bishops score

Bishop score: is a clinical scoring system of the cervix to quantify the degree of ripeness of cervix prior to IOL

	0	1	2	3
Dilatation of cervix (cm)	0	1-2	3-4	5 or more
Consistency of cervix	Firm	Medium	Soft	
Length of cervical canal (cm)	>2	2-1	1-0.5	<0.5
Position of cervix	Posterior	Central	Anterior	
Station(cm above ischial	-3	-2	-1 or 0	Below the spine

High scores >6 indicate a favorable cervix: easier , shorter induction process that results mostly in successful vaginal delivery.

Low scores <6 indicate an unfavorable cervix :longer time with higher risk of failure

Such women need IOL by drugs that induce cervical ripening and uterine contractions at the same time (like PGs)

1.Natural:

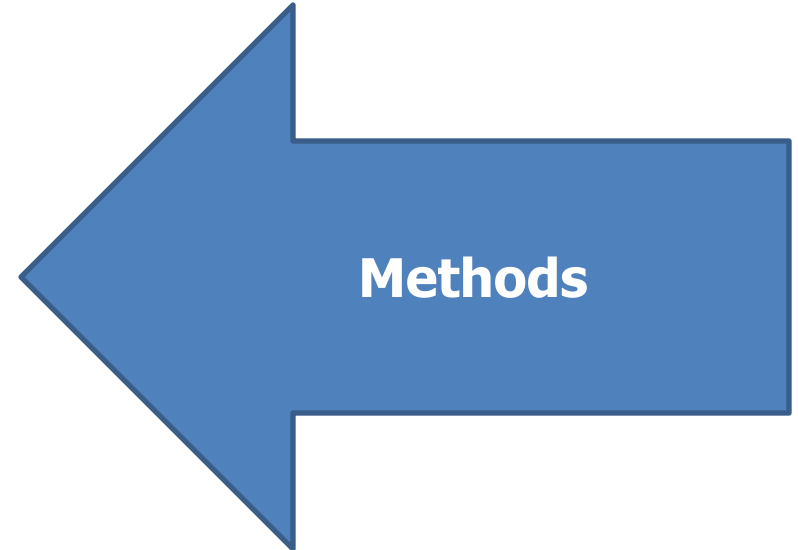
- Sexual intercourse
- Acupuncture/acupressure
- Membrane sweeping

2.Mechanical:

- Ballon catheters.
- Laminaria tents.
- Synthetic osmotic dilators.
- Amniotomy

3.chemical: A: Non-hormonal

- Herbs: evening primrose oil
- Enemas
- Castor oil



B.Hormonal:

- Prostaglandin E2 (dinoprostone)
- Misoprostol (synthetic PGE1 analog)
- Oxytocin
- Mifepristone(anti-progesterone)
- Relaxin
- Nitric oxide donor

Conclusion

- Induction of labor is a critical obstetric intervention aimed at initiating uterine contractions before spontaneous onset, to achieve vaginal delivery when the benefits of early delivery outweigh the risks of continuing pregnancy.

References

1. NICE NG207 — “Inducing labour” (published 4 November 2021).
2. Obstetrics by ten teachers 21st edition